Kelsey McGuire

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Professional Summary

I'm a motivated and personable MSc Student at the University of British Columbia with over 3 years of diverse experience in terrestrial and aquatic environmental sampling, including the collection and processing of soil, water, and vegetation samples. This field work has been supported by a combination of R coding and GIS to solve spatial problems and create public-facing content.

Education

M. Sc., GEOGRAPHY **2024 -**

University of British Columbia (Vancouver, BC)
***Thesis Title**: A little sedge goes a long way: Vegetation as a key driver of littoral Methane flux*
Thesis Advisor(s): Dr. McKenzie Kuhn

B. Sc., GEOGRAPHICAL SCIENCES WITH DISTINCTION **2021 - 2024**

University of British Columbia (Vancouver, BC)
GPA: 3.95/4.33 or 85%; Dean's List Recipient 2021W; 2022W; 2023W

Research Experience

MASTER OF SCIENCE STUDENT | BOREAL-ARCTIC BIOGEOCHEMISTRY LAB (UBC) | 09/01/2025 –

Advisors: Dr. McKenzie Kuhn

- Lead a project to understand the influence of aquatic vegetation on Northern boreal lake Carbon budgets within Kaska ancestral territory (British Columbia) through field obtained measurements on methane and carbon dioxide flux, vegetation presence, and other water quality variables (i.e., pH, temperature).
- I analyzed a dataset of over 300 captured data points, where data was quality controlled for outliers, filtered based on relevant conditions and then modelled and displayed using R code and QGIS. Disseminated this information in a graduate thesis and reports to Indigenous partners.

UNDERGRADUATE RESEARCH ASSISTANT | GEOSPATIAL ECOLOGY LAB (UBC) | 09/01/2023– 08/31/2024

Advisors: Dr. Naomi Schwartz

- Lead a project with the objective of understanding belowground processes in a forest-savanna mosaic of Cambodia through analyses of fine roots and soil organic carbon.
- Study included the collection of 499 soil samples, pre-processing soil for laboratory analyses, and statistical analyses on results identifying patterns in fine-root biomass and soil organic carbon presence.
- Aided in capturing geospatial data in the field using a Trimble R780 and Trimble Connect to catalog sampling site locations and the creation of study maps from this data.

UNDERGRADUATE RESEARCH ASSISTANT | UBC MICROMETEOROLOGY LAB (UBC) |05/03/2023 – 03/31/2024

Advisors: Dr. Sara H. Knox, Zoe (Hehan) Zhang (MSc Student)

- Aided in fieldwork, where assistance was given to take GHG measurements at 48 sample sites within a post-fire peatland bog ecosystem.
- Coded R Scripts for various tasks including:
  - Automating the process of calculating Vegetation Indices from PhenoCam images using the Phenopix package in combination with other R packages.
  - Adapted a Shiny app from Ameriflux to suite the needs of the Micromet lab which included filtering through primary and secondary data, and plotting the requested data based on user input.

Relevant Courses

STAT545 | INTRODUCTORY DATA ANALYSIS | UNIVERSITY OF BRITISH COLUMBIA

- Focused on efficiently employing functions within R's tidyverse and ggplot2 packages. Main tasks included learning to properly filter, clean, and present data based on the structure and goals of the given objective.
- Key outputs included a shared GitHub project that employed R Markdowns and displayed the use of many tidyverse functions including filter(), select(), mutate(), alongside a Shiny App that was created to visualize methane data against environmental variables based on a user supplied dataset.

GEOS370 (ADVANCED GEOGRAPHIC INFORMATION SCIENCE); GEOS373 (INTRODUCTORY REMOTE SENSING)

- Completed multiple individual and group projects using geospatial software, such as:
  - Used the suitability modeler within ArcGIS Pro, along with Kriging and Nearest Neighbour techniques, to analyse and identify suitable Caribou Habitats in Canada's North that was presented through StoryMaps. Automated portions of this analysis using ArcGIS' built-in python notebook and workflows.
  - Used ENVI and Google Earth Engine to conduct numerous analyses on satellite imagery, where collection, pre- and post-processing analyses were performed to compare the effectiveness of supervised versus unsupervised classifiers in identifying sea level rise surrounding New Orleans, LA.

Technical Skills & Certifications

Programming	*Software*	*Field Instrumentation*	*Certifications*
• **R** (Proficient) • **GitHub** (Proficient) • **Python** (Intermediate)	• **GIS (Arc/Q)** (Proficient) • **Microsoft Office** (Proficient) • **ENVI** (intermediate) • **Google Earth Engine** (Beginner)	• **LICOR** GHG analyzers • **Floating Chambers, Bubble Traps** (carbon flux monitoring) • **Water Sampling** (Hanna Probe, YSI)	• Remote First Aid, Workplace First Aid, Transport Endorsement • **Class 5** License

Scholarships & Awards

WALL RESEARCH AWARD **$25 000 (JUNE 2025)**

NORTHERN SCIENTIFIC TRAINING PROGRAM **$3300 (MAY 2025)**

NSERC CANADA GRADUATE SCHOLARSHIP – MASTER'S (CGS-M) **$27 000 (APR. 2025)**

WCS WESTON FAMILY BOREAL RESEARCH FELLOWSHIP **$10 000 (MAR. 2025)**

BRITISH COLUMBIA GRADUATE SCHOLARSHIP **$17 500 (MAY 2024)**

NSERC UNDERGRADUATE STUDENT RESEARCH AWARD (USRA) **$6000 (MAY 2024)**

UBC GEOGRAPHY ALUMNI SCHOLARSHIP **$3450 (NOV. 2023)**

Memberships & Extracurriculars

ASLO STUDENT MEMBER **2025/2026**

CGU STUDENT MEMBER **2025/2026**